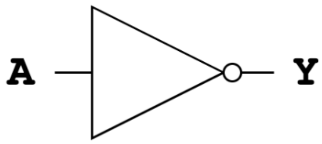
Digital Logic Overview

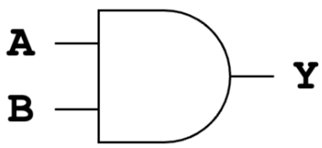
Inverter/NOT gate



$$Y = \overline{A}$$

A	Y
0	1
1	0

AND Gate



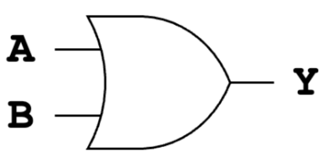
$$Y = A \cdot B$$

or

$$Y = AB$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate



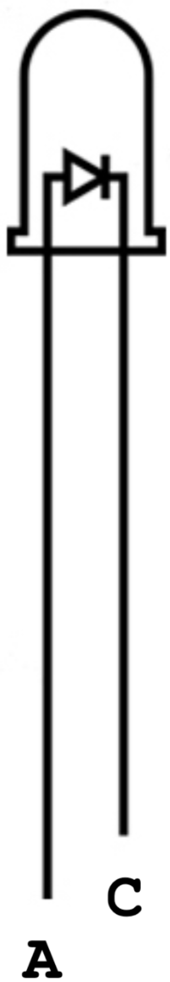
$$Y = A + B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Component Reference

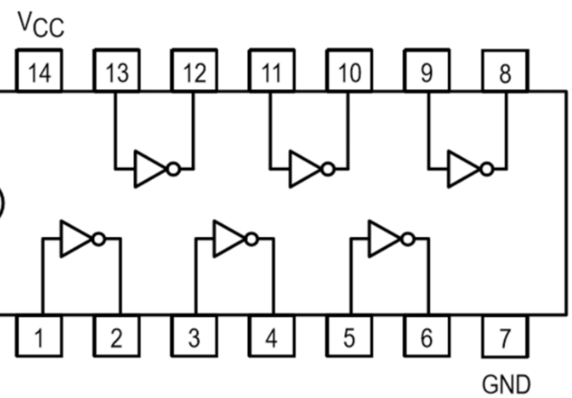
LED

(Light Emitting Diode)

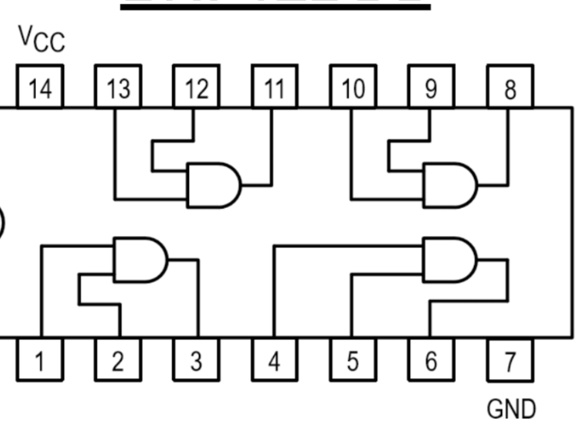


When a sufficiently high voltage is applied to the anode (A), relative to the cathode (C), the LED will allow current to flow and light up. To light up the LED, apply +5V to the anode, and 0V to the cathode

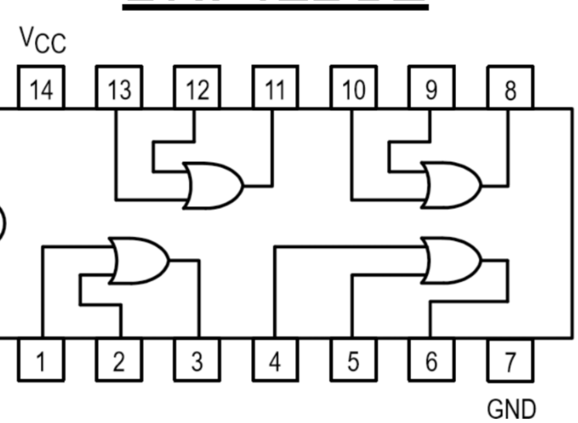
SN74LS04



SN74LS08

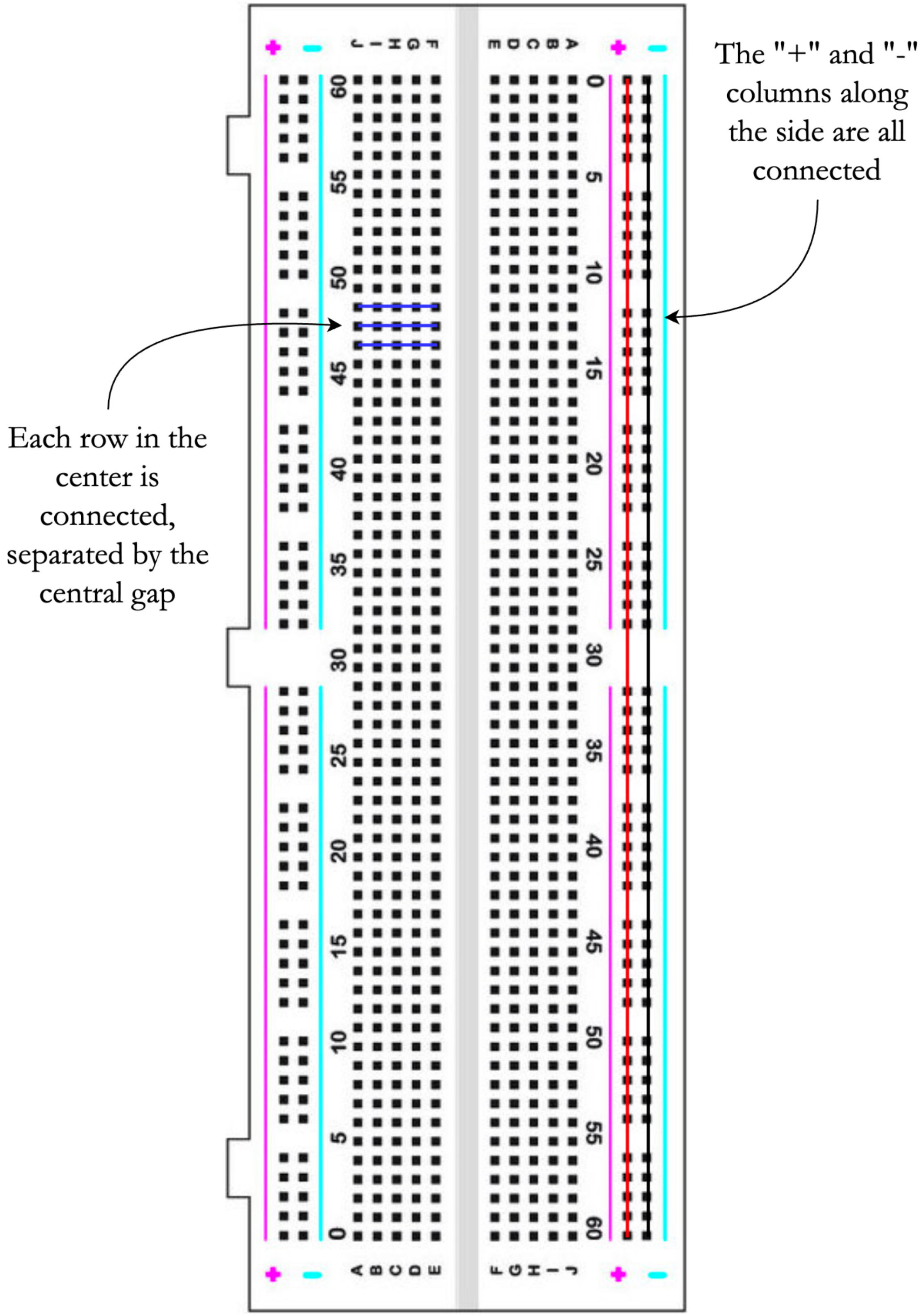


SN74LS32



$V_{CC} = +5V$ $GND = 0V$

Breadboard Connections



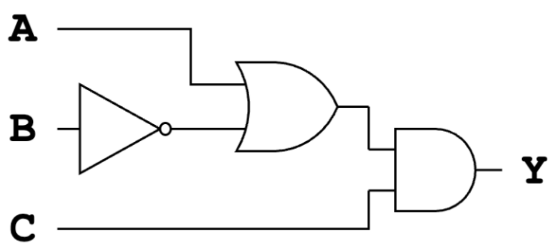
Activities

Task 1: Draw the logic gate diagram corresponding to the following logic equation. Additionally, fill in the truth table on the right

$$Y = (AB) + \overline{C}$$

A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Task 2: Determine the logic equation from the given diagram. Additionally, fill in the truth table on the right



A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Exercise 1: Connect the LED on your breadboard to light it up!

Tips:
- The long lead (anode) should be connected to +5V (the red power rail)
- The short lead (cathode) should be connected to 0V (the blue power rail)
- Avoid using any rows that are already used

Exercise 2: Connect the LED using your logic gates such that when you have the input at +5V, the LED is off, and when you have the input at 0V, the LED is on. In other words:

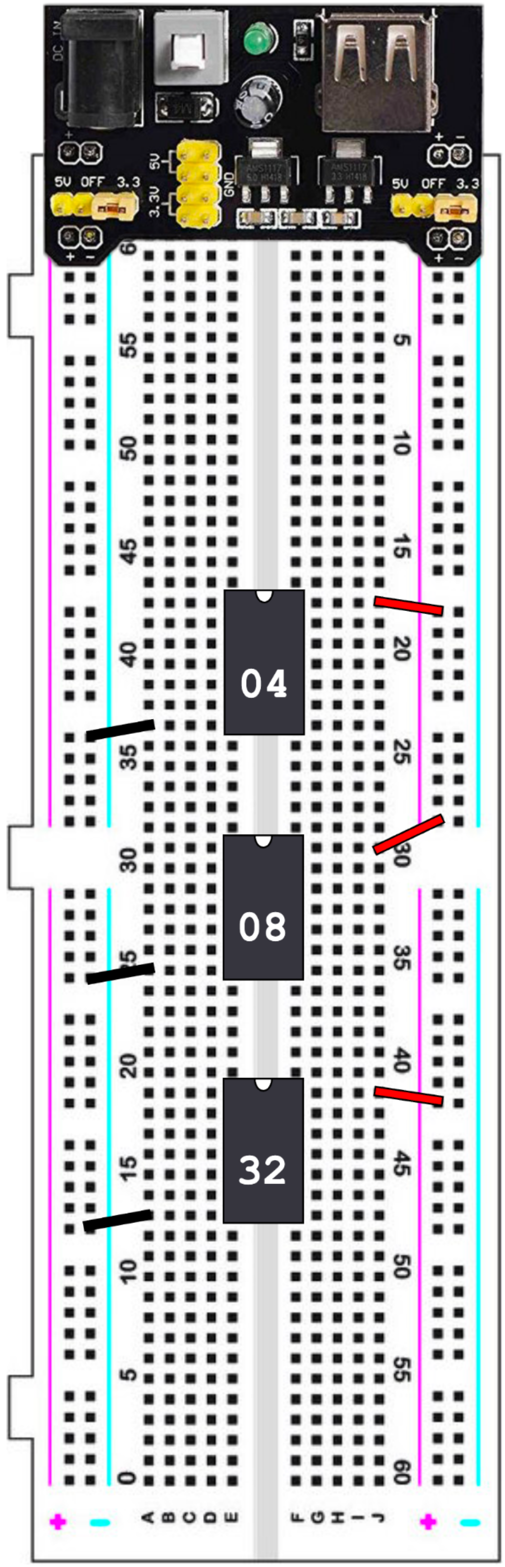
$$Y = \overline{A}$$

Tips:
- The LED's cathode should always be connected to ground (0V)
- The LED's anode should be connected to the output of one of your digital logic gates
- The input to the system should be an input to one of your logic gates

Exercise 3: Connect the LED using your logic gates to achieve the logic equation from Task 1. Vary the inputs; is the output consistent with what your truth table expects?

Tips:
- The LED's cathode should always be connected to ground (0V)
- The LED's anode should be connected to the output of one of your digital logic gates
- The LED being "on" corresponds to an output of 1, whereas the LED being "off" corresponds to an output of 0
- You will need to use all three types of logic gates

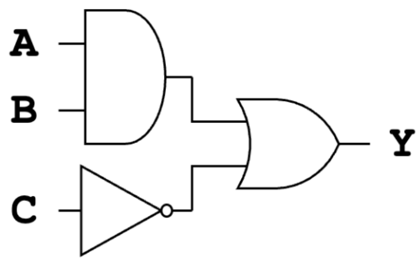
Starting Setup



Solutions:

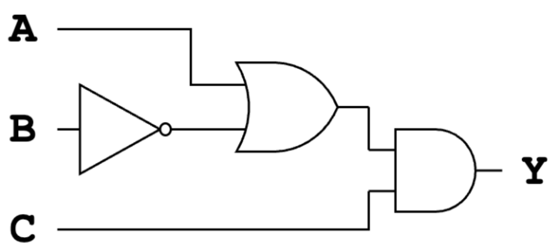
Task 1: Draw the logic gate diagram corresponding to the following logic equation. Additionally, fill in the truth table on the right

$$Y = (AB) + \overline{C}$$



A	B	C	Y
0	0	0	**1**
0	0	1	**0**
0	1	0	**1**
0	1	1	**0**
1	0	0	**1**
1	0	1	**0**
1	1	0	**1**
1	1	1	**1**

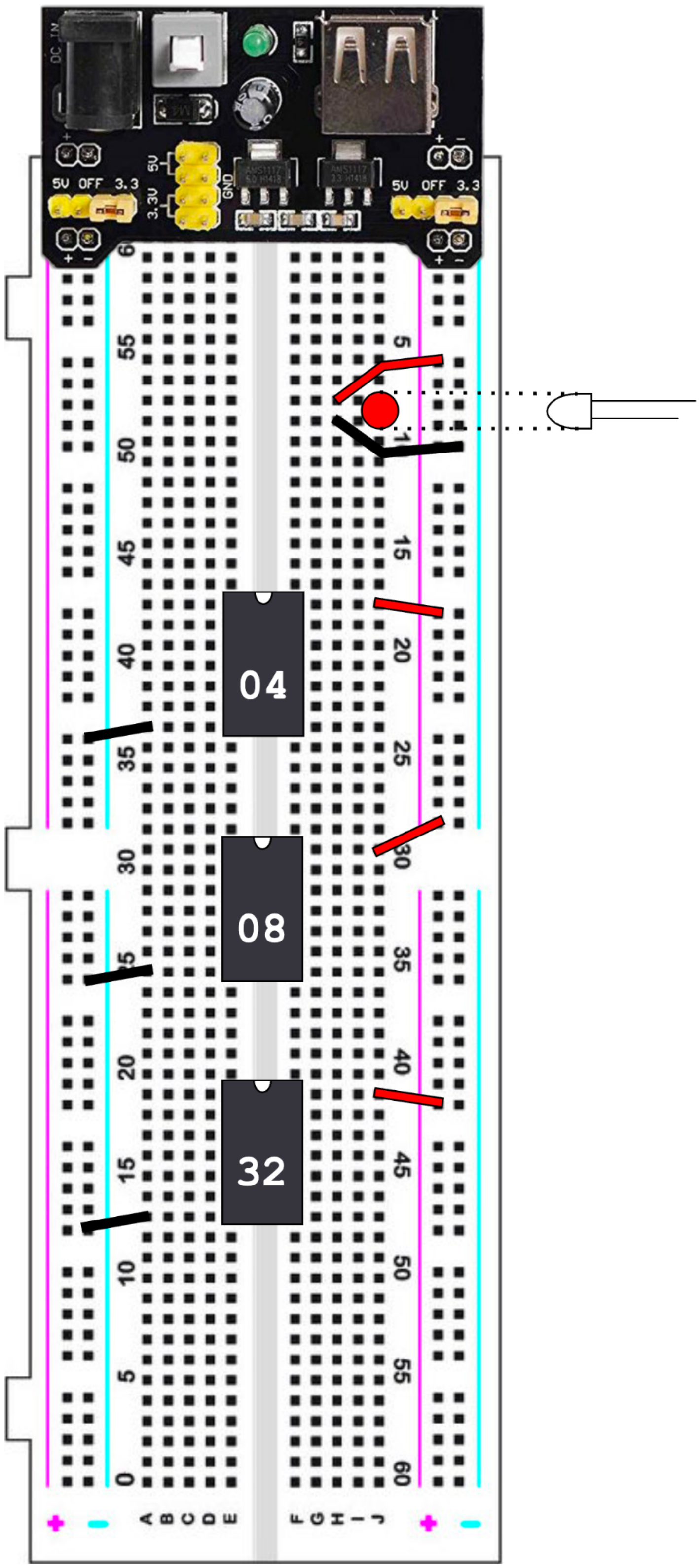
Task 2: Determine the logic equation from the given diagram. Additionally, fill in the truth table on the right



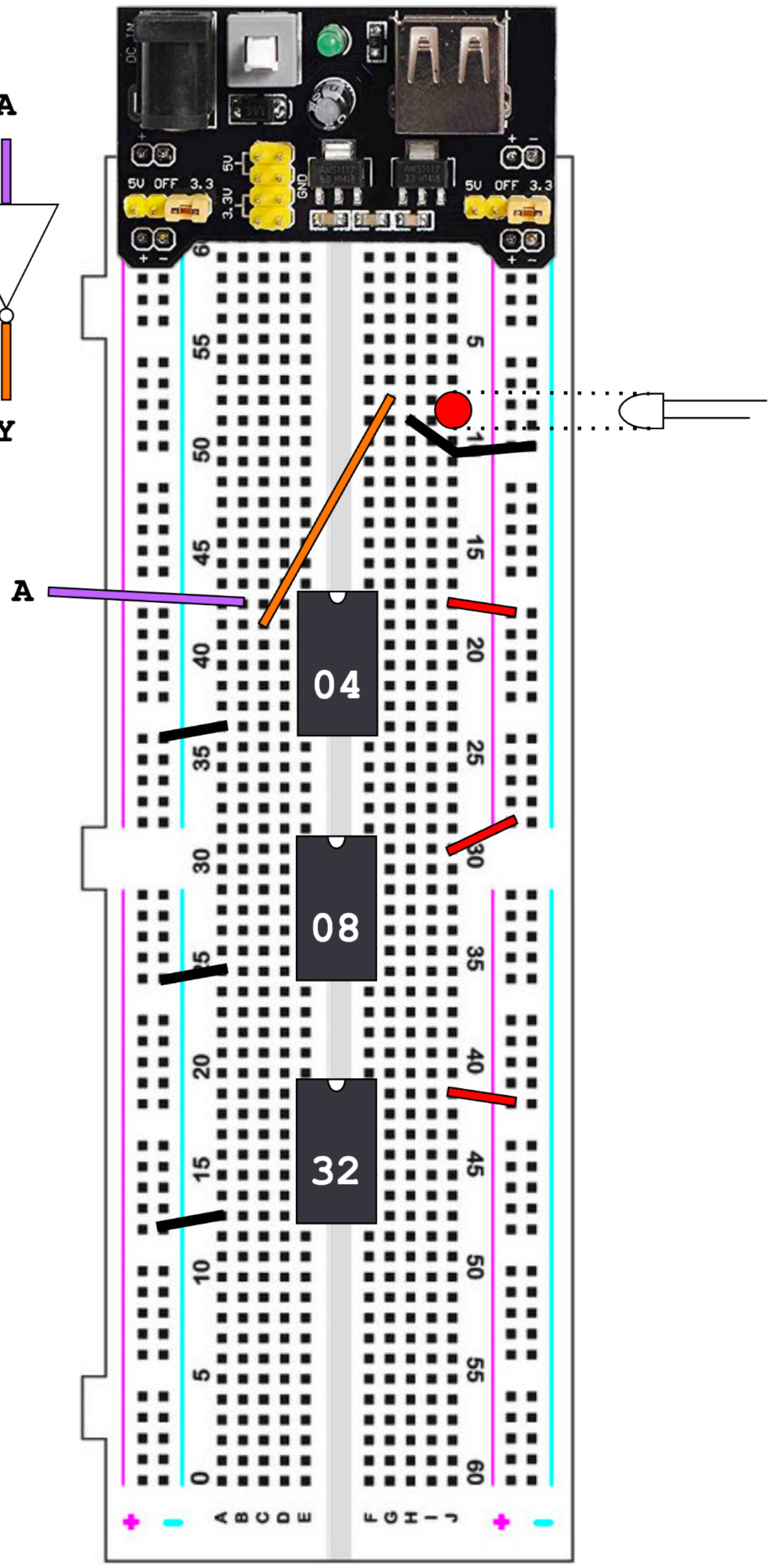
$$Y = (A + \overline{B}) \cdot C$$

A	B	C	Y
0	0	0	**0**
0	0	1	**1**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**1**
1	1	0	**0**
1	1	1	**1**

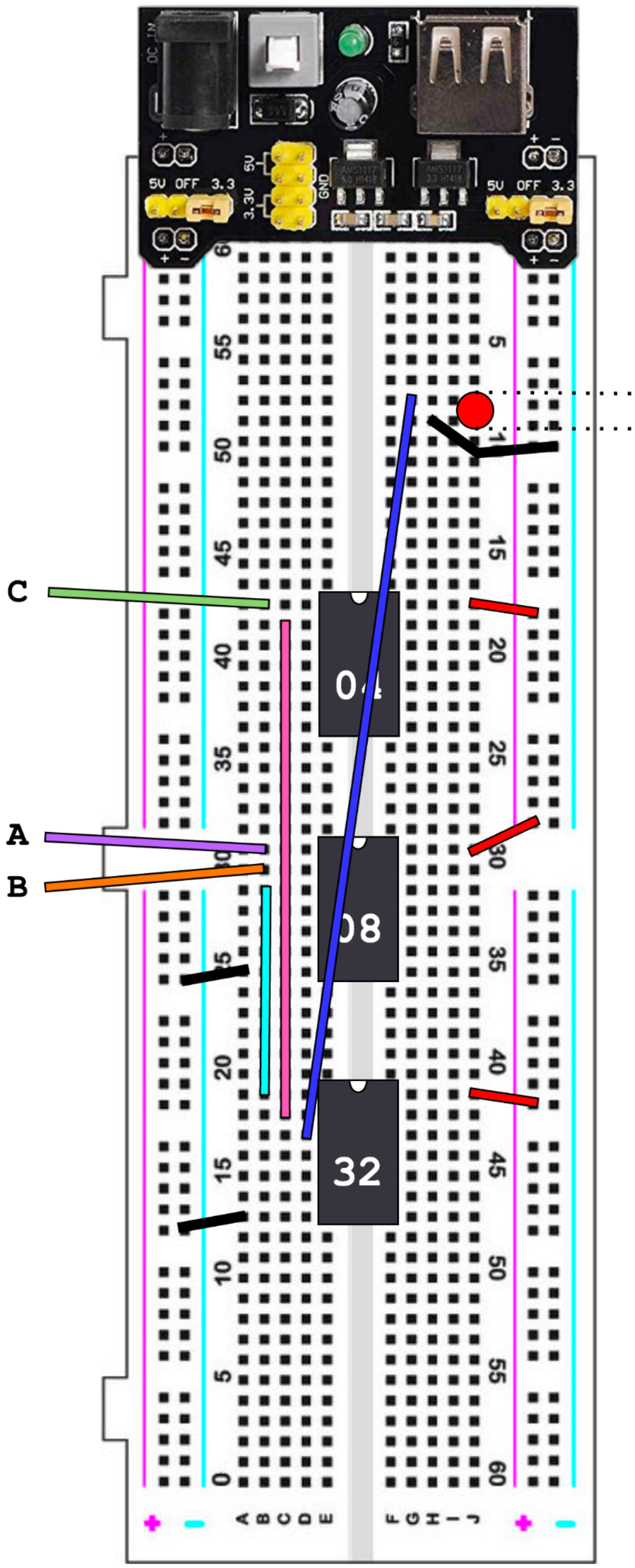
Exercise 1 Solution:

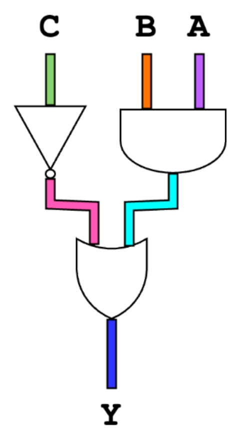


Exercise 2 Solutions:



Exercise 3 Solutions:





Parts List:

- 1x Breadboard Starter Kit (Link)
- 1x 9V Battery
- 1x 9V Battery to Barrel Adapter Cable (Link)
- 1x SN74LS04N 6-Channel Inverter (Link)
- 1x SN74LS08N 4-Channel AND Gate (Link)
- 1x SN74LS32N 4-Channel OR Gate (Link)
- 1x 5V LED (Red/Green)

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